

Series 02 - RStudio

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GitHub Repository

View our GitHub repository <https://github.com/ayeshawaheed9/IEAP-Series02-RStudio>

Team Workflow Strategy Using Git

We decided to create separate branches for each section of the assignment so that we could work independently without interfering with each other's work.

Each subsection was placed in a separate `.qmd` sub-document to make collaboration easier. The sub-documents were organized into folders according to their sections, keeping the project structure clean and easy for both team members to navigate.

We also added each other as reviewers on Pull Requests so that we could review each other's code, identify mistakes, and suggest improvements before merging. Before merging the branches, we rendered the main QMD document to check that all sections worked correctly together.

Challenges Encountered and Lessons Learned

Some additional restructuring was done after the creation of branches so they had to be updated from the main branch before we could continue working on them. This was done by taking a pull from the main branch and merging it in the required branch.

Another challenge was making sure that each sub-document could be rendered both independently and as part of the main document. This required loading the necessary libraries and data within each file and carefully managing the file paths so that they worked correctly in both cases.

During the Pull Request reviews, comments were added to point out issues and areas that needed improvement.

The identified issues were discussed, and the requested changes were made before the Pull Requests were merged. This helped us recheck the code, fix errors, and ensure that the final sections were consistent and worked correctly together.

Section 2: Statistical Tests

2.1 Make clear what is a statistical test, and when to use it.

1. What is a median? A mean?

Median: The middle value of a dataset when the values are arranged in order.

Mean: It is the average of all the values, calculated by adding them together and dividing by the number of values.

Reference: <https://openstax.org/books/introductory-statistics-2e>

2. What is a variance? A standard deviation?

Variance: A measure of how the values spread out around the mean.

Standard deviation: The square root of the variance. It shows the typical spread of the data around the mean.

Reference: <https://openstax.org/books/introductory-statistics-2e>

3. What is a normal distribution?

A **normal distribution** is a symmetric, bell-shaped distribution where most values are close to the mean.

Reference: <https://www.geeksforgeeks.org/data-science/mathematics-probability-distributions-set-3-normal-distribution>

4. What is a statistical test?

A **statistical test** is a method applied to research data to determine the importance of results and assess if those results occurred by chance or not, allowing researchers to draw conclusions.

Reference: <https://www.sciencedirect.com/topics/social-sciences/statistical-test>

5. When to use a statistical test?

We use a statistical test when we want to determine whether an observed result is likely to be due to a real effect or to random variation or by chance.

Reference: <https://www.sciencedirect.com/topics/social-sciences/statistical-test>

6. What is a parametric test? A non-parametric test?

Parametric test: A test that makes assumptions about the distribution of the data, such as normal distribution.

Non-parametric test: A test that makes fewer assumptions and uses medians and rankings for the data.

Reference: <https://builtin.com/data-science/parametric-vs-nonparametric>

7. What are the assumptions of parametric tests?

Common assumptions are:

1. Normality – Data in each group should be normally distributed.
2. Equal Variance – Data in each group should have approximately equal variance.
3. Independence – Data in each group should be randomly and independently sampled from the population.
4. No Outliers – There should be no extreme outliers.

Reference: <https://www.statology.org/parametric-tests-assumptions>

8. What are the assumptions of non-parametric tests?

Non-parametric tests generally require fewer assumptions, but the observations still need to be independent, randomly sampled (collected appropriately) and the data must be suitable for the chosen test.

Reference: <https://www.itl.nist.gov/div898/handbook/prc/section2/prc2.htm>

9. What is a p-value?

A **p-value** tells us how likely the observed result is if there is no real effect.

A small p-value suggests that the result is less likely to be due to chance.

Reference: <https://openstax.org/books/introductory-statistics-2e/pages/9-4-rare-events-the-sample-decision-and-conclusion>

10. What is the risk of error when using a statistical test?

There are two main errors:

- **Type I error:** Rejecting the null hypothesis when it is actually true.
- **Type II error:** Not rejecting the null hypothesis when it is actually false.

Reference: <https://openstax.org/books/introductory-statistics-2e/pages/9-2-outcomes-and-the-type-i-and-type-ii-errors>

11. What is the difference between a paired and an unpaired test?

Paired test: Used when the observations are related, such as measurements taken from the same person before and after an intervention.

Unpaired test: Used when the observations come from two independent groups.

Reference: <https://pmc.ncbi.nlm.nih.gov/articles/PMC2881615/>

2.2 Effect of treatment over time

2.2.1 The data

Loading and exploring the data

```
# Load the tidyverse
library(tidyverse)
```

```
— Attaching core tidyverse packages — tidyverse 2.0.0 —
✓ dplyr      1.2.1    ✓ readr      2.2.0
✓ forcats    1.0.1    ✓ stringr    1.6.0
✓ ggplot2    4.0.3    ✓ tibble     3.3.1
✓ lubridate  1.9.5    ✓ tidyr      1.3.2
✓ purrr      1.2.2

— Conflicts — tidyverse_conflicts() —
* dplyr::filter() masks stats::filter()
* dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
conflicts to become errors
```

```
# Read the data file
prepost <- read.csv("data/PrePost.csv")

# Names of the variables
names(prepost)
```

```
[1] "perf" "time"
```

```
# First rows, summary statistics and number of rows
head(prepost)
```

```
  perf  time
1  125 Before
2  130 Before
3  132 Before
4  135 Before
5  136 Before
6  138 Before
```

```
summary(prepost)
```

```
      perf      time
Min.   :125.0 Length  :16
1st Qu.:132.0 N.unique : 2
Median :136.0 N.blank  : 0
Mean   :136.4 Min.nchar: 5
3rd Qu.:140.2 Max.nchar: 6
Max.   :148.0
```

```
nrow(prepost)
```

```
[1] 16
```

```
# Display all rows to check how the data are ordered
prepost
```

```
      perf  time
1    125 Before
2    130 Before
3    132 Before
4    135 Before
5    136 Before
6    138 Before
7    140 Before
8    145 Before
9    127  After
10   132  After
11   133  After
12   136  After
13   139  After
14   141  After
15   145  After
16   148  After
```

```
# Number of measures at each time point
table(prepost$time)
```

After	Before
8	8

The file contains 16 rows, 8 measures Before and 8 measures After rehabilitation, which corresponds to 8 patients. There is no patient identifier, and within each group the values are sorted in increasing order. **Reference:** data provided by the course instructor on the course Moodle page: <https://moodle.umontpellier.fr/mod/assign/view.php?id=995977>. Analyses were performed with R (R Core Team, 2026): <https://doi.org/10.32614/R.manuals> and the tidyverse (Wickham et al., 2019): <https://doi.org/10.21105/joss.01686>

What are the measured variables, and what do they measure?

The file contains two variables. “perf” is a performance measure taken on patients with cardiac conditions, but the data file does not specify what it measures exactly. “time” indicates when the measure was taken, before or after the rehabilitation training.

What are the units?

The unit of “perf” is not specified in the data file. “time” is a categorical variable and has no unit.

What are the possible values?

The observed values of “perf” range from 125 to 148. “time” takes two values, Before and After.

What are the expected changes over rehabilitation?

Assuming that higher values of perf mean better performance, perf is expected to increase after rehabilitation. Since the nature of perf is not specified, this expected change is an assumption.

Summary table

Variable	Description	Unit	Possible values	Expected change
perf	Performance measure (exact nature not specified in the data file)	not specified	observed 125 to 148	increase after rehabilitation, assuming higher values mean better performance
time	Moment of the measurement	none	Before, After	not applicable

2.2.2 The analysis

Does the treatment have an effect?

Yes, Perf increased significantly after rehabilitation (Wilcoxon signed-rank test, $p < 0.05$), assuming that the rows are ordered by patient. The comparison, the test and the graph supporting this answer are detailed below.

What comparison are you going to make?

The Before and After values were measured on the same 8 patients, so we compared perf Before and After rehabilitation with a paired comparison. The data file contains no patient identifier, so we assumed that the rows are in the same patient order in both groups.

Which test will you use? Parametric or non-parametric?

For a paired comparison, normality is checked on the differences between the two measures of each patient.

```
# Separate the Before and After values
before <- prepost$perf[prepost$time == "Before"]
after  <- prepost$perf[prepost$time == "After"]

# Shapiro-Wilk test on the differences (After minus Before)
shapiro.test(after - before)
```

Shapiro-Wilk normality test

```
data:  after - before
W = 0.89896, p-value = 0.2828
```

Normality of the differences was not rejected ($p > 0.05$), but with only 8 patients the Shapiro-Wilk test has very low power, so we used the non-parametric Wilcoxon signed-rank test.

```
# Wilcoxon signed-rank test (paired, non-parametric)
wilcox.test(after, before, paired = TRUE)
```

Wilcoxon signed rank exact test

```
data:  after and before
V = 36, p-value = 0.007812
alternative hypothesis: true location shift is not equal to 0
```

Which graph illustrates the effect of the treatment?

A line plot connecting the two measures of each patient shows the individual change, which suits a paired design.

```
# Patient identifier, assuming the same patient order in both groups
prepost$patient <- rep(1:8, times = 2)
```

```
# Order the time axis: Before then After
prepost$time <- factor(prepost$time, levels = c("Before", "After"))

# One line per patient, from Before to After
ggplot(prepost, aes(x = time, y = perf, group = patient)) +
  geom_line() +
  geom_point() +
  labs(title = "Perf before and after rehabilitation",
       x = "Time",
       y = "Perf (unit not specified)") +
  theme_bw()
```

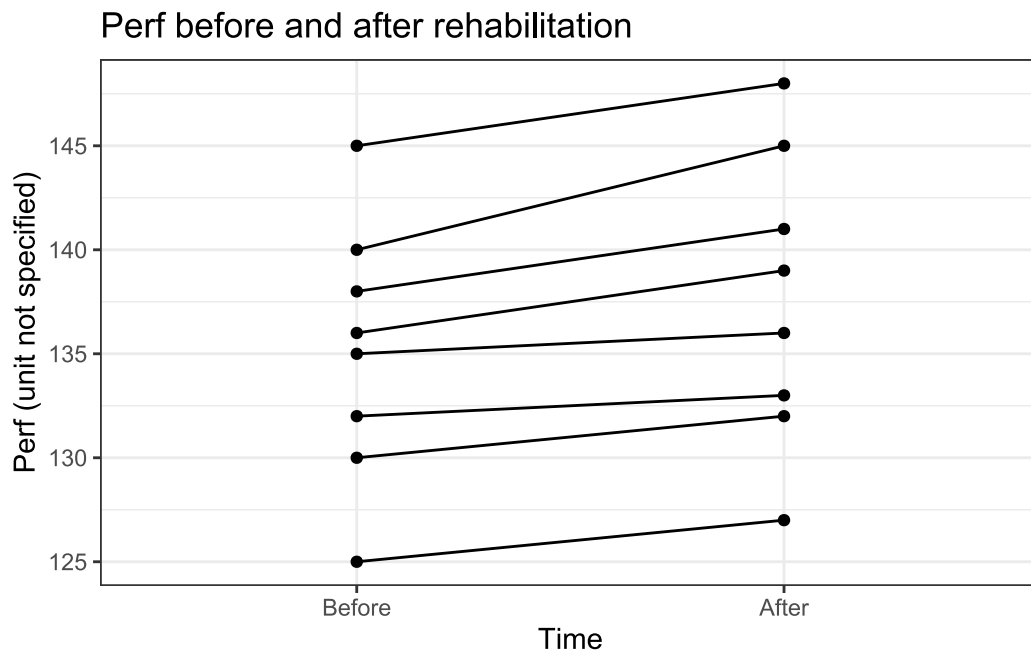


Figure: Perf of each patient before and after rehabilitation. Each line connects the two measures of the same patient, assuming that the rows are ordered by patient. All 8 patients show an increase, ranging from 1 to 5 units.

What sentence will you write in your thesis to explain your result?

After the rehabilitation program, perf increased significantly in the 8 patients, with a median increase of 2.5 units (Wilcoxon signed-rank test, $p < 0.05$), assuming that the rows of the data file are ordered by patient.

This conclusion relies on the assumption that the rows are in the same patient order in both groups. Since the values of each group are sorted in increasing order, this pairing could not be verified, and sorting alone would produce positive differences. The original data with a patient

identifier would be needed to confirm this result. Moreover, since the data only include measures before and after the program, we cannot assess whether the effect is lasting.

2.3 Testing some stereotypes

2.3.1 The data

Loading the data

```
# Load the tidyverse
library(tidyverse)

# Data path, works alone or from the main document
data_dir <- if (dir.exists("data")) "data" else "../..data"
```

```
# Look at the raw file to identify the separator and the header
readLines(file.path(data_dir, "snore.txt"), n = 5)
```

```
[1] "age\tweight\theight\talcohol\tsex\tsnore\ttabac"
[2] "47\t71\t158\t0\tH\tN\t0"
[3] "56\t58\t164\t7\tH\t0\tN"
[4] "46\t116\t208\t3\tH\tN\t0"
[5] "70\t96\t186\t3\tH\tN\t0"
```

```
# Read the tab-separated file
snore <- read.delim(file.path(data_dir, "snore.txt"))

# Type of each variable and summary statistics
str(snore)
```

```
'data.frame':  100 obs. of  7 variables:
 $ age    : int  47 56 46 70 51 46 40 46 49 39 ...
 $ weight: int  71 58 116 96 91 98 112 77 76 119 ...
 $ height: int 158 164 208 186 195 188 193 165 164 196 ...
 $ alcohol: int  0 7 3 3 2 0 5 0 0 3 ...
 $ sex    : chr  "H" "H" "H" "H" ...
 $ snore  : chr  "N" "0" "N" "N" ...
 $ tabac  : chr  "0" "N" "0" "0" ...
```

```
summary(snore)
```

age	weight	height	alcohol
Min. :23.00	Min. : 42.00	Min. :158.0	Min. : 0.00
1st Qu.:43.00	1st Qu.: 77.00	1st Qu.:166.0	1st Qu.: 0.00
Median :52.00	Median : 95.00	Median :186.0	Median : 2.00
Mean :52.27	Mean : 90.41	Mean :181.1	Mean : 2.95
3rd Qu.:62.25	3rd Qu.:107.00	3rd Qu.:194.0	3rd Qu.: 4.25
Max. :74.00	Max. :120.00	Max. :208.0	Max. :15.00

sex	snore	tabac
Length :100	Length :100	Length :100
N.unique : 2	N.unique : 2	N.unique : 2
N.blank : 0	N.blank : 0	N.blank : 0
Min.nchar: 1	Min.nchar: 1	Min.nchar: 1
Max.nchar: 1	Max.nchar: 1	Max.nchar: 1

```
# Number of people in each category
table(snore$sex)
```

```
F H
25 75
```

```
table(snore$snore)
```

```
N O
65 35
```

```
table(snore$tabac)
```

```
N O
36 64
```

Description of the data

The file contains 100 people and 7 variables, with no missing values: age (years), weight (kg), height (cm), alcohol (unit not specified), sex (H = man, F = woman), snore (O = yes, N = no) and tabac (O = smoker, N = non-smoker). Units of weight and height are not given but were deduced from the range of values. The sample includes 75 men and 25 women, 35 snorers and 64 smokers.

Reference: data provided by the course instructor on the course Moodle page: <https://moodle.umontpellier.fr/mod/assign/view.php?id=995977>

Measure of corpulence

Since weight depends on height, corpulence was assessed with the body mass index (BMI = weight / height²). **Reference:** World Health Organization (2000): <https://apps.who.int/iris/handle/10665/42330>

```
# BMI in kg/m², with height converted from cm to m
snore$bmi <- snore$weight / (snore$height / 100)^2
summary(snore$bmi)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
16.20	27.56	28.07	27.22	28.38	32.55

2.3.2 The analysis

```
# Libraries, data and BMI, needed when rendered alone
library(tidyverse)
data_dir <- if (dir.exists("data")) "data" else "../..data"
snore <- read.delim(file.path(data_dir, "snore.txt"))
snore$bmi <- snore$weight / (snore$height / 100)^2
```

Do the data confirm the following stereotypes?

Are snorers fatter?

```
# Normality of BMI in each group
shapiro.test(snore$bmi[snore$snore == "0"])
```

Shapiro-Wilk normality test

```
data: snore$bmi[snore$snore == "0"]
W = 0.72636, p-value = 9.81e-07
```

```
shapiro.test(snore$bmi[snore$snore == "N"])
```

Shapiro-Wilk normality test

```
data: snore$bmi[snore$snore == "N"]  
W = 0.78486, p-value = 2.323e-08
```

Normality of BMI was rejected in both groups ($p < 0.05$), so snorers and non-snorers were compared with the Mann-Whitney test. Since the distributions had a similar shape in both groups (see figure below), the test was interpreted as a comparison of medians.

```
# Median BMI per group and Mann-Whitney test  
tapply(snore$bmi, snore$snore, median)
```

```
      N      0  
28.06892 28.07632
```

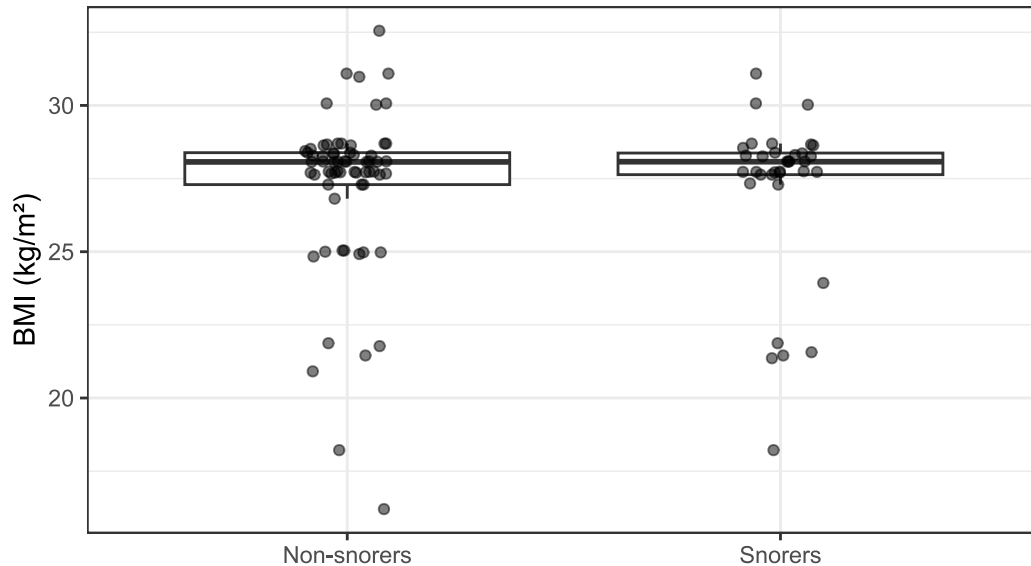
```
wilcox.test(bmi ~ snore, data = snore)
```

Wilcoxon rank sum test with continuity correction

```
data: bmi by snore  
W = 1128.5, p-value = 0.951  
alternative hypothesis: true location shift is not equal to 0
```

```
# Boxplot of BMI by snoring status  
ggplot(snore, aes(x = snore, y = bmi)) +  
  geom_boxplot(outlier.shape = NA) +  
  geom_jitter(width = 0.1, height = 0, alpha = 0.5) +  
  scale_x_discrete(labels = c("N" = "Non-snorers", "0" = "Snorers")) +  
  labs(title = "BMI of snorers and non-snorers", x = "", y = "BMI (kg/m²)") +  
  theme_bw()
```

BMI of snorers and non-snorers



Snorers did not have a higher BMI than non-snorers (median 28.1 kg/m² in both groups, Mann-Whitney test, $p > 0.05$). The data do not confirm the stereotype that snorers are fatter.

Do snorers drink or smoke more?

```
# Normality of alcohol consumption in each group  
shapiro.test(snore$alcohol[snore$snore == "0"])
```

Shapiro-Wilk normality test

```
data:  snore$alcohol[snore$snore == "0"]  
W = 0.90387, p-value = 0.005032
```

```
shapiro.test(snore$alcohol[snore$snore == "N"])
```

Shapiro-Wilk normality test

```
data:  snore$alcohol[snore$snore == "N"]  
W = 0.76256, p-value = 6.818e-09
```

Normality of alcohol consumption was rejected in both groups ($p < 0.05$), so snorers and non-snorers were compared with the Mann-Whitney test.

```
# Median alcohol consumption per group and Mann-Whitney test
tapply(snore$alcohol, snore$snore, median)
```

```
N 0
0 4
```

```
wilcox.test(alcohol ~ snore, data = snore)
```

Wilcoxon rank sum test with continuity correction

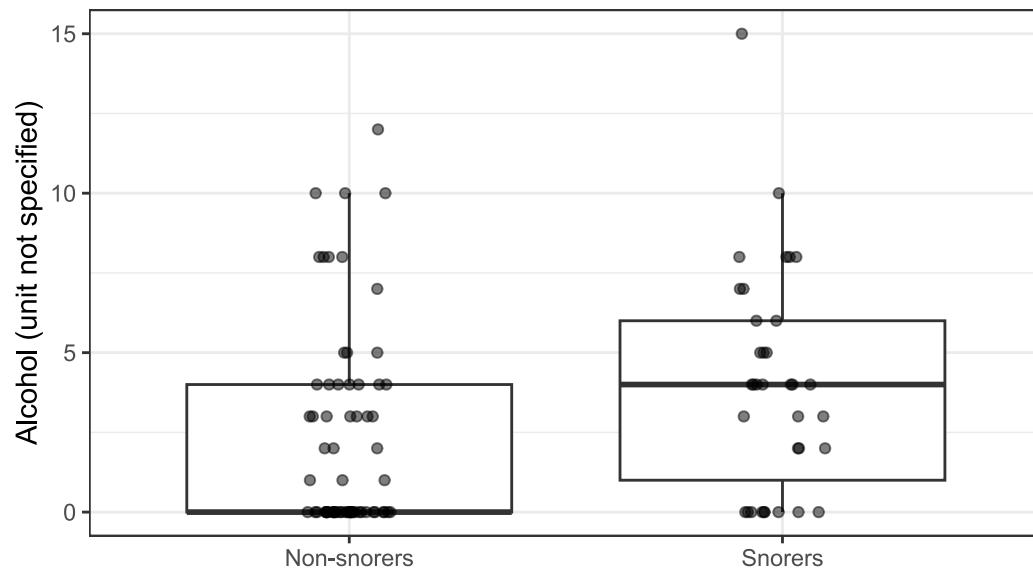
data: alcohol by snore

W = 788, p-value = 0.008594

alternative hypothesis: true location shift is not equal to 0

```
# Boxplot of alcohol consumption by snoring status
ggplot(snore, aes(x = snore, y = alcohol)) +
  geom_boxplot(outlier.shape = NA) +
  geom_jitter(width = 0.1, height = 0, alpha = 0.5) +
  scale_x_discrete(labels = c("N" = "Non-snorers", "0" = "Snorers")) +
  labs(title = "Alcohol consumption of snorers and non-snorers", x = "", y =
"Alcohol (unit not specified)") +
  theme_bw()
```

Alcohol consumption of snorers and non-snorers



Snorers drank significantly more alcohol than non-snorers (median 4 vs 0, Mann-Whitney test, $p < 0.05$). The data confirm the stereotype that snorers drink more.

Since both variables are categorical, smoking was compared between snorers and non-snorers with a chi-squared test, after checking that all expected counts were at least 5.

```
# Contingency table, row percentages and chi-squared test
tab_snore <- table(snore$snore, snore$tabac, dnn = c("snore", "tabac"))
tab_snore
```

```
      tabac
snore  N   0
N 21 44
0 15 20
```

```
round(prop.table(tab_snore, margin = 1) * 100, 1)
```

```
      tabac
snore   N    0
N 32.3 67.7
0 42.9 57.1
```

```
test_snore <- chisq.test(tab_snore)
test_snore$expected
```

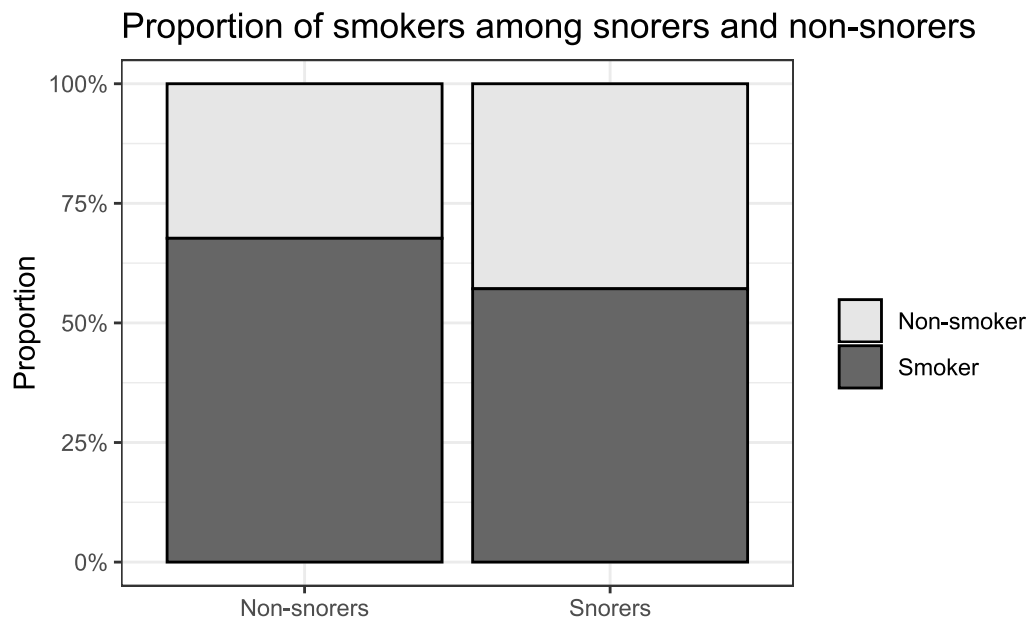
```
      tabac
snore   N    0
N 23.4 41.6
0 12.6 22.4
```

```
test_snore
```

Pearson's Chi-squared test with Yates' continuity correction

```
data:  tab_snore
X-squared = 0.68872, df = 1, p-value = 0.4066
```

```
# Proportion of smokers by snoring status
ggplot(snore, aes(x = snore, fill = tabac)) +
  geom_bar(position = "fill", color = "black") +
  scale_x_discrete(labels = c("N" = "Non-snorers", "0" = "Snorers")) +
  scale_fill_grey(start = 0.9, end = 0.4, labels = c("N" = "Non-smoker", "0" =
"Smoker")) +
  scale_y_continuous(labels = scales::percent) +
  labs(title = "Proportion of smokers among snorers and non-snorers", x = "", y
= "Proportion", fill = "") +
  theme_bw()
```



Snorers did not smoke more than non-snorers (57.1% vs 67.7% of smokers, chi-squared test, $p > 0.05$). The data do not confirm the stereotype that snorers smoke more.

Are men fatter?

```
# Normality of BMI in each sex  
shapiro.test(snore$bmi[snore$sex == "H"])
```

Shapiro-Wilk normality test

```
data: snore$bmi[snore$sex == "H"]  
W = 0.78184, p-value = 3.456e-09
```

```
shapiro.test(snore$bmi[snore$sex == "F"])
```

Shapiro-Wilk normality test

```
data: snore$bmi[snore$sex == "F"]  
W = 0.71462, p-value = 1.154e-05
```

Normality of BMI was rejected in both groups ($p < 0.05$), so men and women were compared with the Mann-Whitney test.

```
# Median BMI per sex and Mann-Whitney test
tapply(snore$bmi, snore$sex, median)
```

```
      F      H
27.72748 28.07504
```

```
wilcox.test(bmi ~ sex, data = snore)
```

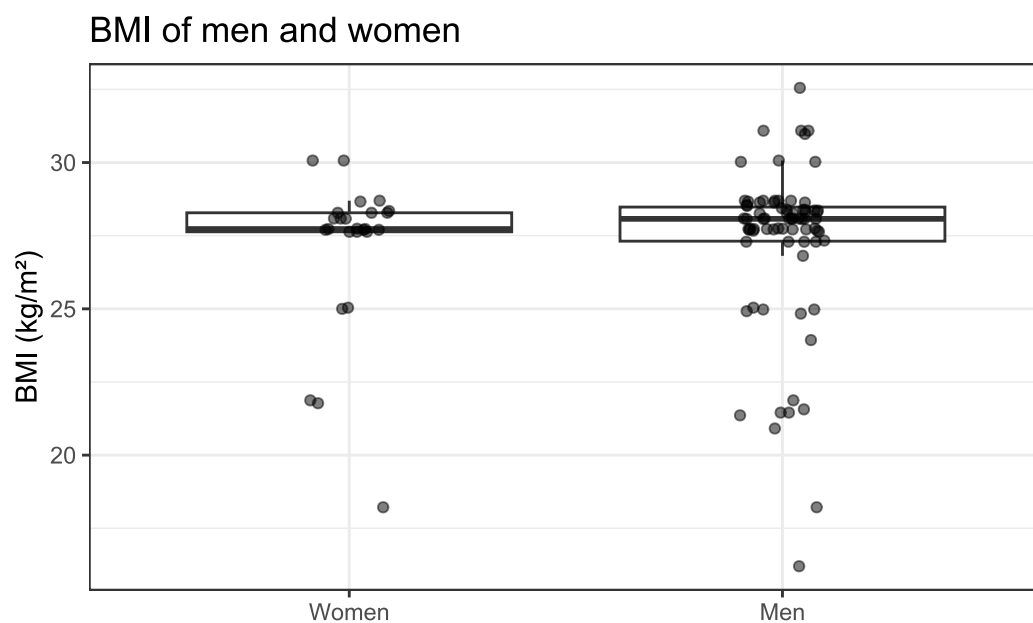
Wilcoxon rank sum test with continuity correction

data: bmi by sex

W = 850, p-value = 0.4884

alternative hypothesis: true location shift is not equal to 0

```
# Boxplot of BMI by sex
ggplot(snore, aes(x = sex, y = bmi)) +
  geom_boxplot(outlier.shape = NA) +
  geom_jitter(width = 0.1, height = 0, alpha = 0.5) +
  scale_x_discrete(labels = c("F" = "Women", "H" = "Men")) +
  labs(title = "BMI of men and women", x = "", y = "BMI (kg/m²)") +
  theme_bw()
```



Men did not have a higher BMI than women (median 28.1 vs 27.7 kg/m², Mann-Whitney test, $p > 0.05$). The data do not confirm the stereotype that men are fatter.

Do women smoke less?

Since both variables are categorical, smoking was compared between men and women with a chi-squared test, after checking that all expected counts were at least 5.

```
# Contingency table, row percentages and chi-squared test
tab_sex <- table(snore$sex, snore$tabac, dnn = c("sex", "tabac"))
tab_sex
```

```
      tabac
sex  N    0
  F 15  10
  H 21  54
```

```
round(prop.table(tab_sex, margin = 1) * 100, 1)
```

```
      tabac
sex  N    0
  F 60  40
  H 28  72
```

```
test_sex <- chisq.test(tab_sex)
test_sex$expected
```

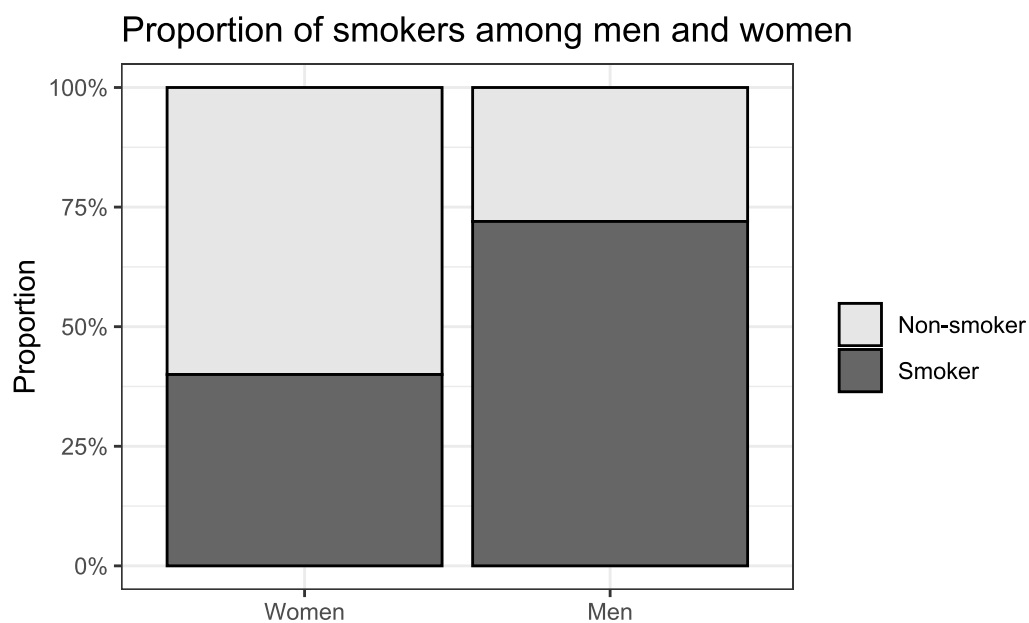
```
      tabac
sex  N    0
  F  9  16
  H 27  48
```

```
test_sex
```

```
Pearson's Chi-squared test with Yates' continuity correction

data:  tab_sex
X-squared = 7.0023, df = 1, p-value = 0.00814
```

```
# Proportion of smokers by sex
ggplot(snore, aes(x = sex, fill = tabac)) +
  geom_bar(position = "fill", color = "black") +
  scale_x_discrete(labels = c("F" = "Women", "H" = "Men")) +
  scale_fill_grey(start = 0.9, end = 0.4, labels = c("N" = "Non-smoker", "0" =
"Smoker")) +
  scale_y_continuous(labels = scales::percent) +
  labs(title = "Proportion of smokers among men and women", x = "", y =
"Proportion", fill = "") +
  theme_bw()
```



Women smoked significantly less than men (40% vs 72% of smokers, chi-squared test, $p < 0.05$). The data confirm the stereotype that women smoke less.

From a more general perspective: are there any correlations between variables?

Since BMI and alcohol consumption were not normally distributed, correlations between the numeric variables were assessed with the Spearman correlation.

```
# Numeric variables only
num <- snore[, c("age", "weight", "height", "alcohol", "bmi")]

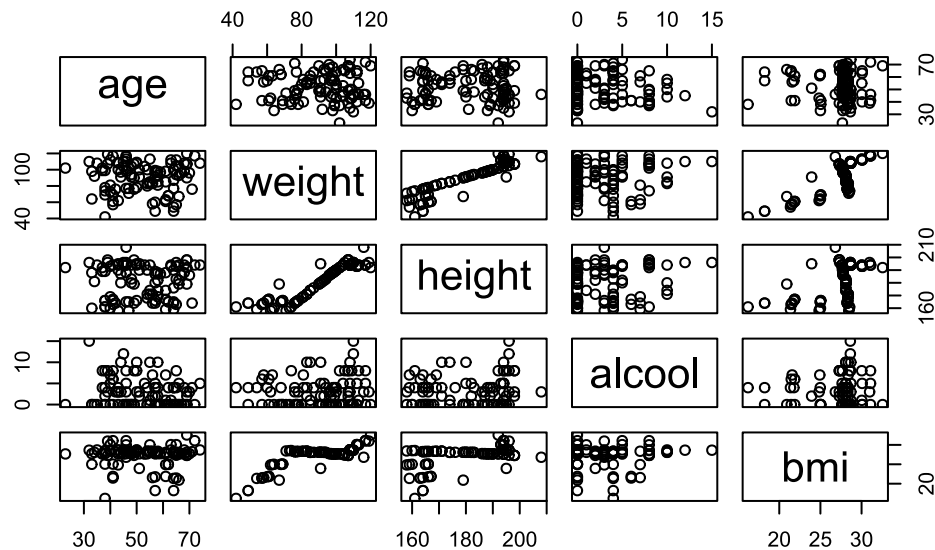
# Spearman correlation matrix
round(cor(num, method = "spearman"), 2)
```

	age	weight	height	alcohol	bmi
age	1.00	-0.01	0.01	-0.11	-0.05
weight	-0.01	1.00	0.93	0.09	0.54
height	0.01	0.93	1.00	0.13	0.30
alcohol	-0.11	0.09	0.13	1.00	0.02
bmi	-0.05	0.54	0.30	0.02	1.00

```
# Rho and p-value for each pair of variables
pairs_list <- combn(names(num), 2)
results <- t(apply(pairs_list, 2, function(v) {
  test <- cor.test(num[[v[1]]], num[[v[2]]], method = "spearman", exact = FALSE)
  c(var1 = v[1], var2 = v[2], rho = round(unname(test$estimate), 2), p =
signif(test$p.value, 3))
}))
results
```

	var1	var2	rho	p
[1,]	"age"	"weight"	"-0.01"	"0.886"
[2,]	"age"	"height"	"0.01"	"0.936"
[3,]	"age"	"alcohol"	"-0.11"	"0.287"
[4,]	"age"	"bmi"	"-0.05"	"0.635"
[5,]	"weight"	"height"	"0.93"	"1.68e-44"
[6,]	"weight"	"alcohol"	"0.09"	"0.397"
[7,]	"weight"	"bmi"	"0.54"	"5.06e-09"
[8,]	"height"	"alcohol"	"0.13"	"0.214"
[9,]	"height"	"bmi"	"0.3"	"0.00243"
[10,]	"alcohol"	"bmi"	"0.02"	"0.835"

```
# Scatter plot of each pair
pairs(num)
```



Spearman correlations between numeric variables showed a very strong positive correlation between weight and height ($\rho = 0.93$, $p < 0.05$). BMI was also correlated with weight ($\rho = 0.54$, $p < 0.05$) and height ($\rho = 0.30$, $p < 0.05$), but these correlations are expected since BMI is computed from both variables. Age and alcohol consumption were not correlated with any other variable ($p > 0.05$). These conclusions remain unchanged after a Bonferroni correction for the 10 tests performed (threshold $p < 0.005$). The almost perfect alignment between weight and height is unusual for human data and suggests that this data set may have been partly simulated.

Grading Criteria Checklist

The following checklist was used to ensure that all grading criteria were met and that our Git workflow was properly followed.

Project Structure

- ☒ Clear and complete [README.md](#) and [LICENSE](#) files included.
- ☒ Complete [IEAP-Series02-Rstudio.qmd](#) file and corresponding PDF included.
- ☒ Clean folder structure with subfolders and sub-documents.
- ☒ Sub-documents used so team members could work on different sections simultaneously.

Git Workflow

- ☒ Separate branches created for different sections.
- ☒ Meaningful commit messages used, with at least one commit per question.
- ☒ Changes pushed regularly to GitHub.
- ☒ Team members reviewed each other's Pull Requests to recheck the work before merging.
- ☒ Final branches and merged code were checked to ensure no work was lost.

Final Check

- ☑ Master QMD was checked after merging all sections.
- ☑ Final document was rendered and checked.
- ☑ All required files and sections were present and correctly structured.

Data sources

The data files `PrePost.csv` and `snore.txt` were provided by the course instructor and downloaded from the course Moodle page (access restricted to enrolled students): <https://moodle.umontpellier.fr/mod/assign/view.php?id=995977>

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